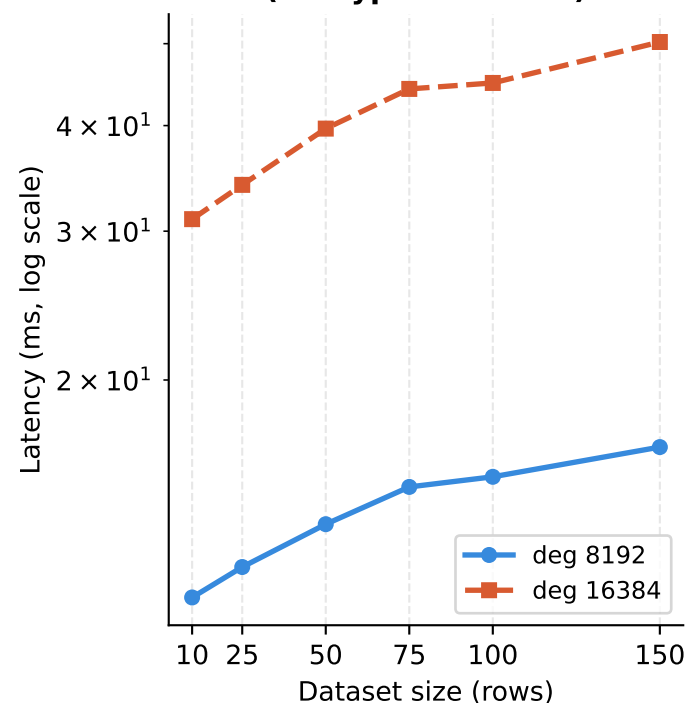


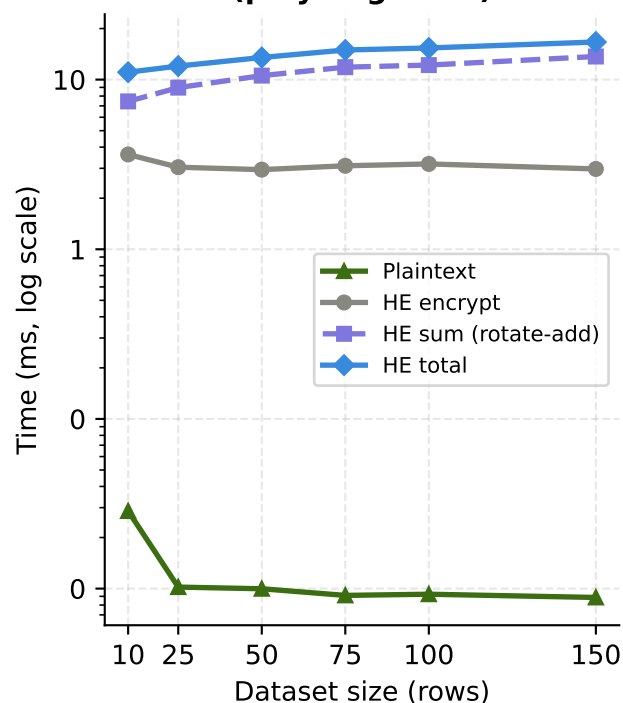
CKKS Homomorphic Encryption: Performance Analysis

Plaintext (NumPy) vs. Encrypted Computation on Iris Dataset

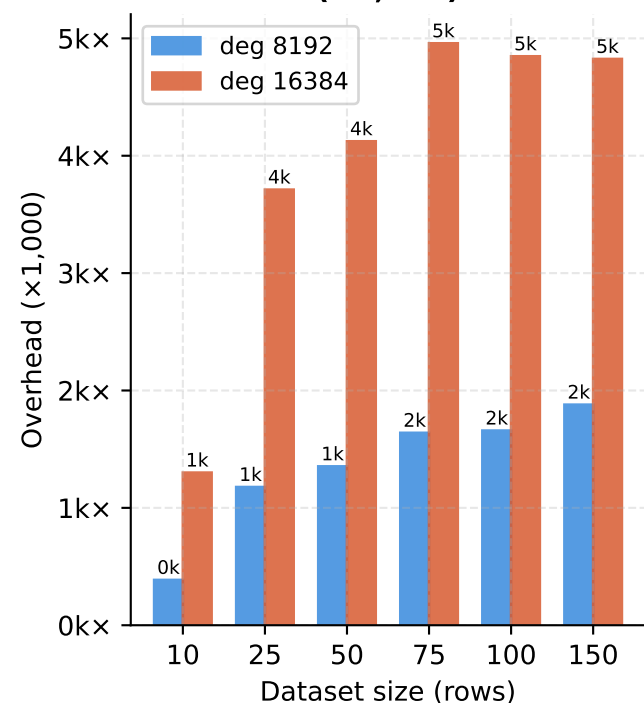
**HE total latency
(encrypt + HE sum)**



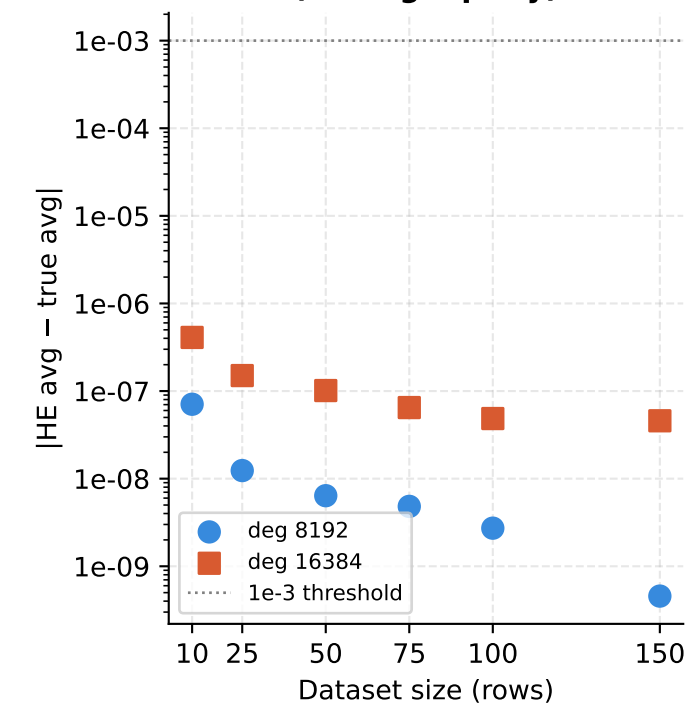
**Plaintext vs. HE components
(poly deg 8192)**



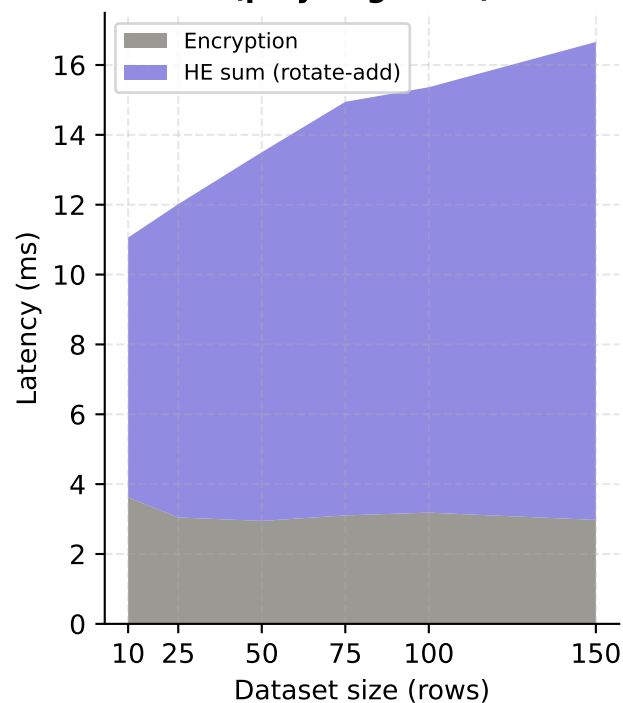
**HE overhead vs. plaintext
($\times 1,000$)**



**CKKS approximation error
(average query)**



**Time breakdown: encrypt vs. sum
(poly deg 8192)**



**Cost of doubling poly degree
(16384 vs. 8192)**

